

From drawing to cut list: a small storage cabinet

Before fabrication, each part must agree with the drawing and cut list. Work through one defined cabinet, check the arithmetic, then apply the same construction to new dimensions.

This lesson teaches fabrication planning. These dimensions are an original teaching example, not a universal standard or permission to operate machinery.

After this lesson you can

- Distinguish outside dimensions from individual panel dimensions.
- Calculate panels fitted between the sides, allowing for panel thickness.
- Calculate body depth when an overlay back is included in the overall depth.
- Prepare a numbered cut list and check quantities, units and assumptions.

Prerequisites: simple arithmetic and dimensions in millimetres. Tools: paper, pencil and a calculator, or the lesson calculator.

Read the brief first

An open-front cabinet with two full-height sides. The top, bottom and middle shelf fit between the sides. A separate back overlays the rear face. Overall depth includes the back. All dimensions are in millimetres.

Lesson assumptions: nominal thickness, butt-joint geometry, no grooves, doors, feet or finishing allowances. Any changed assumption requires a revised drawing and cut list.

W600 × H600 × D300 / t18 / b6 mm

Explanation

Calculate the internal width

Outside width is 600. Each side occupies 18 of that width. The top, bottom and shelf lengths are therefore $600 - 18 - 18 = 564$. This equation applies because those panels sit between the sides in this construction.

$$W - 2t = 600 - 2 \times 18 = 564 \text{ mm}$$

Separate body depth from back thickness

Overall depth is 300, including a 6-thick overlay back. Body panels are $300 - 6 = 294$ deep. The back retains the full outside width and height: 600×600 . A rebated or grooved back would require different geometry.

$$D - b = 300 - 6 = 294 \text{ mm}$$

Check shelf and opening heights

The top, bottom and shelf occupy three thicknesses: $3 \times 18 = 54$. Remaining clear height is $600 - 54 = 546$. Two equal openings are $546 \div 2 = 273$ high each. This is the clear opening height, not the shelf length.

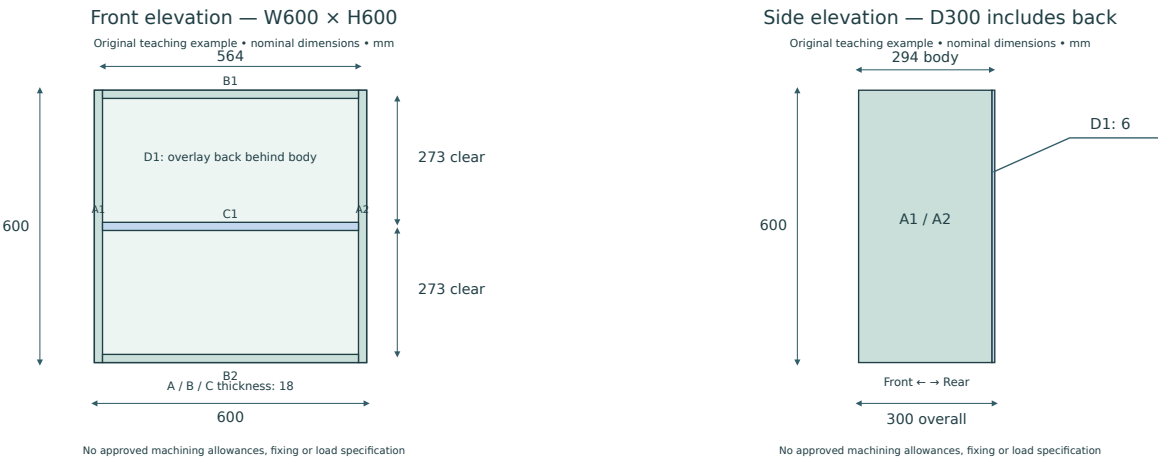
$$(H - 3t) \div 2 = 273 \text{ mm}$$

Turn the drawing into a cut list

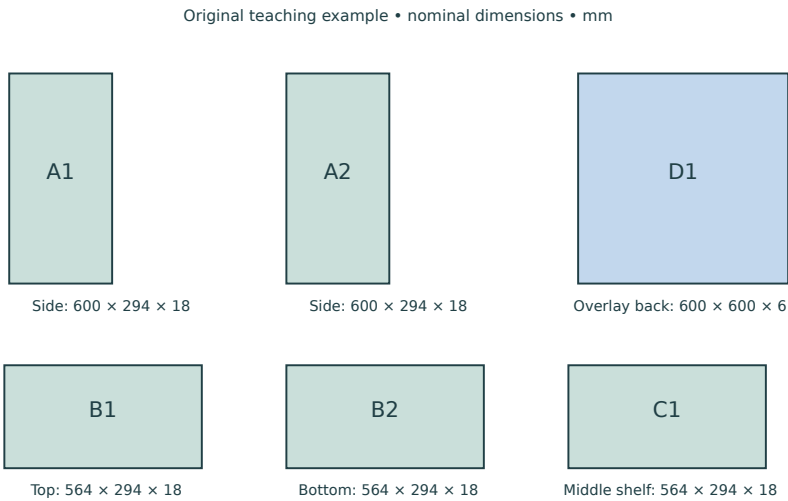
Part	IDs	Qty	Length	Width	Body thickness
Sides	A1 / A2	2	600	294	18
Top & bottom	B1 / B2	2	564	294	18
Middle shelf	C1	1	564	294	18
Overlay back	D1	1	600	600	6

Net panel area alone does not determine stock-board quantity. A cutting layout must also account for material direction, saw kerf and allowances.

Drawings



Part map — 6 pieces, IDs match the cut list



No approved machining allowances, fixing or load specification

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Knowledge check

1. How long is the top fitted between the sides?

- ☐ 600 mm
- ☐ 564 mm
- ☐ 582 mm

2. What is body-panel depth when the overall depth includes the back?

- ☐ 300 mm
- ☐ 306 mm
- ☐ 294 mm

3. How many pieces are in this example?

- ☐ 6
- ☐ 5
- ☐ 7

4. What is required before actual cutting?

- ☐ Approve only the generated picture
- ☐ Review drawings, allowances, material and fixings with a specialist
- ☐ Buy stock from net area alone

Review explanations on page 7 after answering.

Practice & assessment

New task: a cabinet 800 wide × 700 high × 350 deep. Body thickness is 18 and the overlay back is 6. Retain the same construction and two equal openings.

Internal width	Body depth	Clear opening height

Review criteria: consistent drawing/list part IDs; correct dimensions and quantities; declared back construction and units; documented workshop approvals.

The automatic check below assesses arithmetic only. Drawings, part lists and documented assumptions need human review. This pilot does not award a certificate or fabrication competence.

Cost worksheet

Record dated supplier quotes, materials, hardware, labour, finishing and transport. This lesson assumes no market prices.

Item	Quantity / unit	Unit price / quote date	Total
Body stock			
Back stock			
Hardware			
Labour			
Finishing			
Transport			

Review before releasing the list

Match every part ID to the drawing. Check units, quantities, thickness and back construction. Before fabrication, a competent specialist must approve cutting and finishing allowances, material direction, joints, fixing and intended loading. This lesson ends at preparing and checking the planning list.

Reviewer / date / required revisions: _____

Video

Panel placement explanation

A Remotion panel placement explanation using calculated dimensions, in Egyptian Arabic and English. Videos use the platform's existing Bunny delivery. This is a schematic demonstration; the workshop must approve joints, fixings and machining allowances.

[Watch the video on Bunny.](#)

1. Read the brief first

An open-front cabinet with two full-height sides. The top, bottom and middle shelf fit between the sides. A separate back overlays the rear face. Overall depth includes the back. All dimensions are in millimetres.

2. Calculate the internal width

Outside width is 600. Each side occupies 18 of that width. The top, bottom and shelf lengths are therefore $600 - 18 - 18 = 564$. This equation applies because those panels sit between the sides in this construction.

3. Separate body depth from back thickness

Overall depth is 300, including a 6-thick overlay back. Body panels are $300 - 6 = 294$ deep. The back retains the full outside width and height: 600×600 . A rebated or grooved back would require different geometry.

4. Check shelf and opening heights

The top, bottom and shelf occupy three thicknesses: $3 \times 18 = 54$. Remaining clear height is $600 - 54 = 546$. Two equal openings are $546 \div 2 = 273$ high each. This is the clear opening height, not the shelf length.

5. Turn the drawing into a cut list

Record part IDs, quantity, length, width and thickness. Group identical parts in a single row while retaining their IDs. Keep the thinner back as its own item. There are six pieces: five body panels and one back.

6. Review before releasing the list

Match every part ID to the drawing. Check units, quantities, thickness and back construction. Before fabrication, a competent specialist must approve cutting and finishing allowances, material direction, joints, fixing and intended loading. This lesson ends at preparing and checking the planning list.

Lesson guide

Fixed reference answers from this lesson. A connected generative AI assistant is not integrated into this pilot yet.

Why subtract both side thicknesses?

Because the top, bottom and shelf fit between the sides. See the internal-width section; a top extending over the sides would require different geometry.

Is 18 mm a rule for all furniture?

No. It is an assumption for this example only. Material, thickness, joints and loading require technical approval for the particular project.

How do I determine stock-board quantity?

Start with the cut list, then lay out parts on the available stock size with material direction, kerf and allowances. Net area alone is insufficient.

Can I operate a saw after this lesson?

This is a planning lesson. Machine operation requires machine-specific practical instruction, supervision and assessment. See the review section and HSE reference.

Knowledge-check explanations

1. Subtract both side thicknesses: $600 - 2 \times 18 = 564$.
2. The overlay back is included in overall depth: $300 - 6 = 294$.
3. Two sides + top + bottom + shelf + back = six pieces.
4. The teaching list contains no approved machining allowances, board layout, fixing or load specification.

Watching a video or passing the quiz does not establish competence to operate woodworking machinery; practical instruction, supervision and assessment are separate.